ATOMS

1. What are atoms?

.....

.....

2. Atoms are made up of smaller particles called protons, neutrons and electrons. Draw a representation of an atom that is made up of 2 protons, 2 neutrons and 2 electrons (this is a helium atom), showing the position of these three types of particles.

3. Complete the following table:

Sub-atomic particles

	Proton	**Neutron**	**Electron**
Position in an atom			
Charge			

4. In the table below, record the name, the symbol and the number of protons in the following atoms. Also record one interesting fact about each type of atom, and give an example of one object/substance in which the atom occurs.

Atom	**Symbol**	**No. of protons**	**Interesting fact**	**Object in which atom can be found**
hydrogen	H	1	Goes “pop” when lit with a burning match	Water
sodium				
potassium				
magnesium				
calcium				
barium				
carbon				
nitrogen				
oxygen				
sulfur				
fluorine				
chlorine				
bromine				
iodine				
helium				
neon				
copper				
iron				
silver				
mercury				
zinc				

5. To help you learn these symbols of atoms, make up a set of “learning” cards. For each card, on one side write the name of the atom and on the other side write the symbol. You can then test yourself.

6. List the most abundant atoms that are in the following substances or objects:

Substance/Object	**Atoms present** (most abundant)
rocks	
air	
water	
sea water	
bricks	
plastic	
natural gas	
the Sun	
legs of classroom chairs	
glass	
window frames	
paper	
wood	
school dress	
a living organism	
jewellery	
steel	

REVIEW QUESTIONS

7. a) Name the three main sub-atomic particles present in an atom. .....

   b) Which sub-atomic particles are present in the nucleus? .....

   c) Describe where the electron is found in the atom. .....

   d) What are the charges on the three sub-atomic particles? .....

8. Give the symbols of the following atoms

hydrogen	nitrogen
oxygen	helium
chlorine	neon
sodium	calcium
carbon	sulfur
iron	copper
potassium	magnesium
barium	fluorine
bromine	silver
iodine	mercury

9. Give the names of the atoms with the following symbols:

   H

   He

   Hg

   N

   Na

   Ne

F

Fe

Cu

C

Cl

Ca

Ba

Br

K

Mg

S

O

I

Ag

Zn

ELEMENTS

1. **Activity 13**. - Elements - metals and non-metals

Examine each of the elements, carry out the required activities and then record your results in the table.

Element	State at room temperature	Appearance (after cleaning)	Malleable or Brittle ?	Conduction of electricity
Copper				
Sulfur				
Nitrogen				
Magnesium				
Helium				
Zinc				
Neon				
Iron				
Carbon				
Hydrogen				
Oxygen				

Questions

a) Which of the elements in the table are metals?

.....

.....

b) Which of the elements in the table are non-metals?

.....

.....

2. **Activity 14.** - Preparation of two non-metals - hydrogen and oxygen

OXYGEN

i) Pour about 25 mL of hydrogen peroxide into a conical flask.

ii) Add a small amount of manganese dioxide to the hydrogen peroxide and then insert the stopper and bent glass tube into the top of the flask

iii). Collect a test tube full of oxygen by displacement of water

iv) Examine the oxygen and fill out the appropriate column in the table from Activity 13

v) Test the oxygen with a glowing splint. What happened? .....

vi) Draw a diagram of apparatus used to prepare and collect oxygen:

HYDROGEN

i) Pour a small amount of hydrochloric acid into a test tube.

ii) Add several pieces of magnesium to the acid.

iii) Collect the hydrogen gas by placing an empty test tube up-side-down over the mouth of the first test tube.

iv) Examine the hydrogen gas and fill out the appropriate column in the table in Activity 13.

v) Test the hydrogen by putting a burning match at the mouth of the test tube. .....

vi) Draw a diagram of apparatus used to prepare and collect hydrogen

3. Which of the following properties are possessed by metals and which are possessed by non-metals?

Are very good conductors of electricity — Most are solids at room temperature

Most are hard — Normally have a low melting point

Do not conduct electricity — Are malleable

4. **Activity 15.** - Some more elements

Look at the following elements and complete the table:

Name of element	Description of element	Metal or non-metal?
sodium		
calcium		
chlorine		
bromine		
iodine		
silver		
mercury		

5. a) Some pieces of zinc were added to nitric acid and a colourless gas was collected in a test tube. When the gas was lit with a match, a "pop" noise was heard. What was the gas?

.....

b) An orange-coloured solid was placed in a test tube and strongly heated. The solid decomposed to form a silver coloured liquid and a colourless gas. When the gas was tested with a glowing stick, the stick burst into flames. What was the gas?

.....

6. Use your periodic table to identify each of the following elements as a metal or a non-metal:

hydrogen			sodium		potassium
	calcium			carbon	oxygen
fluorine			copper		
bromine		helium			
silver			magnesium		barium
	nitrogen			sulfur	
chlorine		iron			iodine
	neon				mercury

7. Complete the following sentences:

a) Copper is often used to make electrical wires

because .....

b) Saucepans are made from aluminium, iron or copper because metals

are.....

.....

c) Jewellery is made from gold and silver because metals are .....

and .....

8. Complete the following table:

Name of molecule	**Formula of molecule**	**Number and type of atoms present in the molecule**	**"Drawing" of the molecule**
hydrogen molecule	H_2	2 hydrogen atoms	
nitrogen molecule	N_2		
oxygen molecule		2 oxygen atoms	
ozone		3 oxygen atoms	

9. Copper is a solid at room temperature and helium is a gas at room temperature. Which of these two elements has stronger forces of attraction between its atoms?

.....

10. For each of the following elements, briefly describe their appearance, state whether they are a metal or non-metal, where we obtain them and what we use them for:

Name of element	**Description**	**Metal or non-metal?**	**Where obtained**	**Uses**
hydrogen				
helium				
carbon				
nitrogen				
oxygen				
neon				
aluminium				
silicon				
chlorine				
argon				
iron				
copper				
silver				
gold				
mercury				
lead				
uranium				
zinc				

11. a) What is the formula of sodium? .....

b) Neon is a gas made up of single atoms with weak bonds between them.
What is the formula of neon?
.....

c) Chlorine is a yellow-green non-metal element. It is made up of molecules each containing two chlorine atoms bonded together. What is the formula of chlorine?

.....

d) Sulfur is made up of molecules. Each molecule has eight sulfur atoms bonded together. What is the formula of sulfur?

.....

REVIEW QUESTIONS

12. Below are listed some elements together with some of their properties. Identify each of the elements as a metal or a non-metal.

a) Gold - a shiny, golden coloured solid, very good conductor of electricity, very malleable.

.....

b) Cobalt - black-silver solid, very hard, very good conductor of electricity .....

c) Phosphorus - red powdery solid (sometimes a yellow solid), does not conduct electricity

.....

d) Zinc - silver solid, malleable, good conductor of electricity .....

e) Argon - colourless gas .....

f) Iodine - shiny, black solid, does not conduct electricity, brittle .....

13. State why each of the following are made from metals:

a) the reflective backing on a piece of glass to make a

mirror .....

b) electrical wires

.....

c) the element of a toaster

.....

d) foil used for wrapping food

.....

e) the body of a car

.....

14. Write the chemical formulae of the following metals: silver copper

iron sodium magnesium
calcium potassium

15. Identify each of the elements (A - F) that are described below. Make your choice from the following list:
hydrogen, oxygen, bromine, copper, mercury, aluminium

Element A - a brown/gold solid which is a good conductor of electricity .....

Element B - a shiny, silver-coloured liquid .....

Element C - a colourless gas which goes "pop" when a lighted match is placed in it. .....

Element D - a dark red liquid which does not conduct electricity .....

Element E - a shiny silver solid which is very malleable .....

Element F - a colourless gas which will make a glowing splint burst into flame. .....

16. Sketch a representation of each of the following elements, showing the particles of which they are composed:

hydrogen gas (H_2)	helium gas (He)	oxygen gas (O_2)
iron	nitrogen gas (N_2)	neon gas (Ne)

COMPOUNDS

1. **Activity 17** - Ionic and molecular compounds

Examine each of the compounds, carry out the required activities and record your results in the table.

Compound	State at room temperature	Appearance	Melting point (high or low)	Solubility in water	Conductivity of solution
Butter					
Sodium chloride					
Water					
Sugar					
Copper sulfate					
Potassium nitrate					
Wax					
Nickel chloride					
Calcium carbonate					
Carbon dioxide					

Questions

a) Which of the compounds in the table are ionic compounds? .....

.....

.....

b) Which of the compounds in the table are molecular compounds? .....

.....
.....

2. **Activity 18**. - Preparation of a molecular compound - carbon dioxide

i). Add several pieces of calcium carbonate to a test tube .

ii) Pour about 20 mL of hydrochloric acid into the test tube and insert the stopper and bent glass tubing into the top of the test tube

iii) Collect a test tube full of carbon dioxide.

iv). Examine the carbon dioxide and fill out the appropriate column in the table from Activity 17.

v) Bubble some carbon dioxide through some limewater. What happens?

.....
.....

vi) Draw a diagram of apparatus used to prepare and collect carbon dioxide.

3. Identify the following statements as true or false:

a) Water is an example of an ionic compound

b) Ionic compounds have low melting points

c) All molecular compounds are hard and brittle.

d) When sugar dissolves in water it is a good conductor of electricity.

e) Sodium chloride is an ionic compound and so it will be soft and malleable.

f) Butter is a molecular compound and so it will have a high melting point.

g) All ionic compounds are solids at room temperature.

4. Give five examples of some common molecular compounds:

.....

.....

5. The information in the following table refers to molecular compounds. Complete the table.

Name of compound	**Number and type of atoms present in one molecule**	**Formula of compound**
water		
carbon dioxide	one carbon, two oxygens	
hydrogen chloride		HCl
sulfur trioxide	one sulfur, three oxygens	
dinitrogen tetroxide		N_2O_4
sulfuric acid	two hydrogens, one sulfur, four oxygens	
candle wax		$C_{22}H_{46}$

6. Chocolate has a low melting point and salt has a high melting point. Which of these two substances has the stronger attractive forces between its molecules?

.....

.....

7. a) Complete the following diagram showing some molecules in the liquid water and some molecules in gaseous water.



Liquid water
Gaseous water

b) When liquid water is heated to form gaseous water, which bonds are broken, the strong bonds holding the atoms together in the molecules OR the weak bonds between the molecules?

.....

8. Many substances, such as salt (sodium chloride), are made up of ions. What is an ion?

.....

9. Find the names of five substances/objects that are made up of ions.

.....

10. Atoms are made up of protons, neutrons and electrons.

a) Which sub-atomic particle has a +1 charge? .....

b) Which sub-atomic particle has a -1 charge? .....

11. The magnesium ion has 12 protons (positive charges) and 10 electrons (negative charges). It therefore is said to have a +2 charge. It is a positive ion.

The symbol or formula of the magnesium ion is written as Mg^{2+}.

Calculate the charge and give the formulae of the ions in the following table:

Ion	Number of protons	Number of electrons	Charge on the ion	Formula of the ion
sodium ion	11	10		
calcium ion	20	18		
silver ion	47	46		
copper ion	29	27		
iron ion	26	23		

12. Negative ions have more electrons (negative charges) than protons (positive charges). The names of simple negative ions end in "-ide".

For example, the oxide ion is made from an oxygen atom. An oxide ion has 8 protons and 10 electrons. It has a charge of -2 and its formula is written as O^{2-}.

12. (cont) Calculate the charge and give the formulae of the ions in the following table:

Ion	Number of protons	Number of electrons	Charge on the ion	Formula of the ion
chloride ion	17	18		
sulfide ion	16	18		
nitride ion	7	10		
mercury ion	80	76		
bromide ion	35	36		
barium ion	56	54		

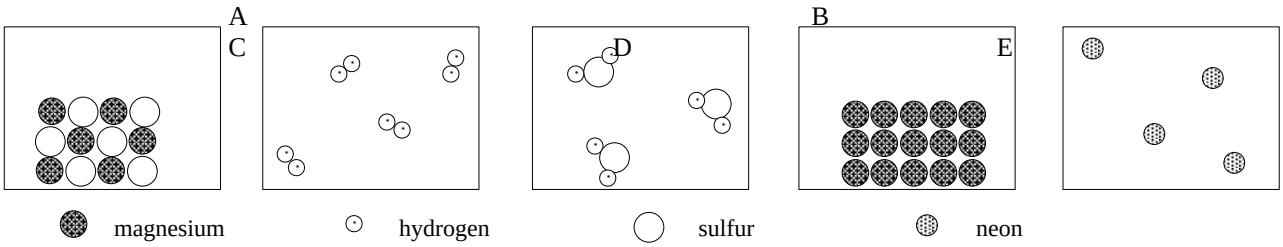
13. Briefly explain why ionic compounds like salt have a high melting point.

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14. Study the following diagrams which represent different types of substances, then answer the questions:



a) Which substances are elements? .....

b) Which substances are compounds? .....

c) Which substances are made up of molecules? .....

d) Which substances are gases? .....

e) What is the state of substance A? .....

f) Which substance is made up of positive and negative ions? .....

g) Which solid substance is a very good conductor of electricity? .....

h) What is the formula of substance B? .....

i) What is the formula of substance C? .....

j) Which substance would have the formula of MgS? .....

REVIEW QUESTIONS

15. Which of the following substances are elements and which are compounds?

nitrogen gas water sodium chloride

helium copper carbon dioxide

magnesium oxide hydrogen gas hydrogen chloride

magnesium ammonia (NH_3) iron

16.. Identify each of the following compounds as an ionic compound or a molecular compound:

Compound A - a blue solid which has a high melting point and is very brittle

Compound B - a white solid which is very hard and brittle and which conducts electricity when it is dissolved in water.

Compound C - a colourless gas which smells like rotten eggs.

Compound D - a yellow solid which melts at a low temperature and is soft.

17. For each of the following substances draw a representation showing the particles (atoms, molecules or ions) that are present:

water	carbon dioxide (CO_2) (a molecular compound)	sodium chloride (NaCl) (an ionic compound)
magnesium oxide (MgO) (an ionic compound)	hydrogen gas (H_2)	helium gas (He)
ammonia gas (NH_3) (a molecular compound)	oxygen gas (O_2)	potassium bromide (KBr) (an ionic compound)
copper	calcium sulfide (CaS) (an ionic compound)	sulfur dioxide gas (SO_2) (a molecular compound)

CHEMICAL REACTIONS

1. **Activity 19.** - Chemical reactions

Mix/react the chemicals as shown in the table below and complete the columns in the table.

Chemicals	Observations	Did a chemical reaction occur?
Add solid sodium carbonate to hydrochloric acid		
Mix solutions of sodium carbonate and copper sulfate		
Mix solutions of sodium chloride and hydrochloric acid		
Mix solutions of copper carbonate and hydrochloric acid		
Mix solutions of nickel chloride and sodium carbonate		
Mix solutions of sodium carbonate and potassium chloride		
Add magnesium to sulfuric acid		
Mix solutions of potassium iodide and lead nitrate		
React magnesium with oxygen (burn a piece of magnesium)		

2. Give five examples of commonly occurring reactions. For each reactions name at least one of the reactants and one of the products.

Reaction	Reactant/s	Product/s

3. For each of the following chemical reactions that are described below
   i) list the reactants and the products
   ii) write a word equation to represent the reaction

a) When mercuric oxide (an orange solid) is heated, a colourless gas called oxygen and a silver liquid called mercury are formed.

reactant: .....

products: .....

word equation: .....

b) When sulfur, a yellow solid, reacts with oxygen, a pungent smelling gas called sulfur dioxide is formed.

reactants: .....

product : .....

word equation: .....

c) When hot sodium is dropped into a jar containing chlorine gas, a violent reaction occurs and a white solid called sodium chloride is formed.

reactants: .....

product: .....

word equation: .....